



EUROSTAT MIGRATION

Visualizing Migration Dynamics Among European Countries

December 18, 2025

Shalva BUTKHUZI, Cesar VILCHERREZ ZARATE



1

INTRODUCTION

Migration is a central demographic and socio-economic phenomenon in Europe, shaping labor markets, population structures, and public policy. While official statistical institutions such as Eurostat provide comprehensive migration data, these datasets are often complex, multidimensional, and difficult to interpret using traditional tabular representations. As a result, understanding migration dynamics across countries, time, and demographic groups remains challenging for non-expert users and even for analysts.

The objective of this project is to design and implement an interactive visualization system that enables exploratory analysis of immigration and emigration flows among European countries. The visualization focuses on revealing spatial, temporal, and demographic patterns in migration data, allowing users to examine how migration volumes evolve over time, how flows differ between countries, and how demographic attributes such as sex and age contribute to these movements.

To achieve this goal, the project combines animated flow-based visualizations with statistical charts and interactive controls. Users can select a country of interest, switch between immigration and emigration perspectives, and explore migration patterns over time using a temporal slider and animation controls. Additional contextual information, including economic indicators and demographic breakdowns, is provided through an integrated statistics panel.

The system is implemented as a web-based application using the D3.js library and is hosted as a GitHub Pages website at <https://shbutkhuzi.github.io/eurostat-vis/> [1]. By leveraging interactive visual encoding techniques, the project aims to support intuitive exploration of large-scale migration datasets and to facilitate a deeper understanding of European migration dynamics.

2

DATA OVERVIEW

This section describes the datasets used in the project, their sources, and the preprocessing steps applied before analysis.

The migration data used in this project were obtained from the Eurostat Database [2], which provides a wide range of statistical datasets covering European countries, including population dynamics, education, industry, transport, and economic indicators. For this project, migration datasets were selected for visualization, and additional economic statistics were collected to complement the analysis of migration dynamics among European countries. The following subsections describe these datasets in detail.

2.1 MIGRATION

The migration datasets used in this project consist of immigration [3] and emigration [4] statistics. These datasets describe migration flows between countries by year, age group, sex, and country of previous or next residence.

In this context, *usual residents* are defined as individuals who have lived in a given country for at least 12 months before the reference time, or who arrived with the intention of staying for at least 12 months [5]. An *immigrant* is defined as a person who arrives to live in a country for a period of at least 12 months, having previously resided in another country. Similarly, an *emigrant* is a person who leaves their country of usual residence for at least 12 months [5]. It should be noted that for some countries, including Finland and Sweden, the statistical definition of emigration is based on a six-month criterion for certain destination countries; however, the resulting impact on the data is reported as negligible [5].

Regarding data accuracy, the number of emigrants recorded by country A moving to country B should, in principle, correspond to the number of immigrants from country A reported by country B. In practice, discrepancies exist due to differences in national registration systems and data collection methodologies. The metadata description of these datasets contains the list of sources for each member country [5]. Additionally, the datasets include flags indicating breaks in time series, provisional values, estimates, or methodological changes. In this project, these flags are ignored due to the simplified analytical approach, and migration trends over time are treated as a single coherent time series.

The datasets provide migration volumes in absolute numbers, representing the number of people registered as immigrants or emigrants within a given year, disaggregated by sex and age group. For some countries and years, information on sex or age group is missing; this issue is handled at the visualization stage, as discussed later in the paper.

The migration data cover the period from 1998 to 2023. Age is grouped in five-year intervals, resulting in a total of 18 age groups. The age definition used in the retrieved dataset follows the *reach* convention, meaning that an individual's recorded age corresponds to the age attained at the end of the reference year. The data were obtained in CSV format via the Eurostat API, and unnecessary columns were removed to reduce dataset size and improve loading performance. The data retrieval and preprocessing steps are documented in the project repository [6].

2.2 GDP

GDP values are used in the project [7] to provide economic context for migration patterns rather than to establish causal relationships. In particular, GDP data support comparisons between the selected country and its main migration partner countries, helping users explore potential associations between economic scale and migration intensity. For each year, countries are ranked by GDP, and the top ten countries are displayed in a horizontal bar chart alongside the selected country when applicable.

Although salary-related indicators were initially considered, their temporal and geographic

availability was more limited and inconsistent across countries. As a result, GDP was selected as the primary economic indicator due to its broader coverage and reliability. This design choice ensures that economic context is available for most countries and years represented in the migration visualization.

2.3 COUNTRY & BORDER GEOJSON

The project dataset also includes GeoJSON files used for map construction. Specifically, it contains country and border cultural vector data at a 1:50m scale, obtained from the Natural Earth dataset [8]. The GeoJSON files were generated by converting the original shapefiles provided by Natural Earth.

Both the country and border GeoJSON files are included in the project repository under the `/data` directory [6]. The decision to store borders as a separate dataset was motivated by visualization requirements. When country geometries are rendered as individual SVG paths, the drawing order determines which elements appear on top. As a result, when emphasizing a country on hover (e.g., by increasing stroke width), parts of its borders may be partially obscured by neighboring countries rendered later in the SVG layer. By rendering borders as a separate GeoJSON layer placed above country fills, country outlines remain consistently visible and can be emphasized uniformly, regardless of neighboring geometries.

2.4 DATA LOADING

All datasets described above are preprocessed and included in the repository, eliminating the need for real-time data retrieval from external sources. The preprocessing is intentionally lightweight: the migration data obtained from the Eurostat database are already well structured, and no external preprocessing tools or libraries (such as Python-based pipelines) are used. The only transformation applied is the removal of unnecessary columns, which is performed using a Bash script included in the project repository. This script is the same one that documents and automates the data retrieval process from the Eurostat database, ensuring transparency and reproducibility.

The datasets are loaded in parallel when the page initializes, reducing overall load time. Using D3's `csv` and `json` methods, the data are filtered, converted to appropriate types, and grouped during the loading process. This preprocessing ensures efficient access and enables faster interactions in subsequent visualizations.

In addition, a one-time validation is performed on the datasets to ensure consistent country mapping between Eurostat data and GeoJSON. The Eurostat dataset is also checked for completeness regarding available age groups and sex. These validation steps are executed during code development and are not repeated on every page load, since once the data have been verified and fixed, repeated checks are unnecessary.

3

MIGRATION FLOW

3.1 VISUAL OVERVIEW

The main landing page presents an interactive political world map that allows users to zoom, pan, and select individual countries. By default, the map is centered on Europe. Whenever the selected country or year changes, the view automatically adjusts to focus on the participating countries, improving the visual perception of migration dynamics. Selectable countries are highlighted using a lighter color to distinguish them from non-participating regions. Once a country is selected, the corresponding migration flows are displayed.

Users can switch between immigration and emigration using a radio button located in the bottom-right corner of the interface. If the selected migration mode is not available for the chosen country or year, the corresponding option is disabled to indicate that the visualization cannot be generated. Restricting the interface to a single active migration mode at a time helps reduce visual clutter and maintains readability.

Migration flows between countries are represented as connections between country centers, defined by label coordinates from the GeoJSON dataset. This choice produces visually balanced and aesthetically pleasing flow layouts. Each flow is animated using particles that move along the connection, with particle speed logarithmically proportional to the migration volume. The speed scale is consistent across countries, years, and migration modes, allowing users to visually compare relative migration intensities regardless of the selected context. The direction of particle movement indicates the direction of migration.

Color is used to encode gender information whenever it is available. Each flow is divided into two segments, where the relative lengths of the red and blue portions represent the proportions of female and male migrants. If gender-disaggregated data are unavailable for a given country or year, the entire flow is rendered in purple. In some cases, data are reported for only one gender; in such situations, the flow appears in a single uniform color. This design enables intuitive interpretation of gender dominance and facilitates comparison between partner countries.

In addition to selecting a country, users can hover over other countries on the map to highlight and contrast their migration relationships. This interaction allows direct comparison of immigration and emigration dynamics between the selected country and a country on which mouse is hovering, with respect to remaining countries for a given year, as illustrated in Figure 1.

The interface also includes a year slider positioned at the bottom center of the page. Users may manually select a specific year or activate a play function that automatically iterates through all available years for the selected country, enabling exploration of migration dynamics over time.

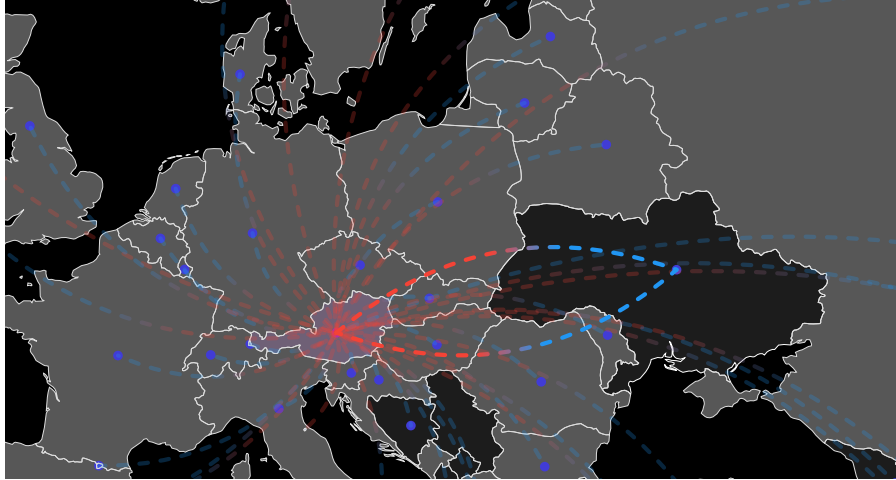


Figure 1: Highlighted migration flows of focused countries within the overall migration network

3.2 IMPLEMENTATION DETAILS

The map and migration flow network are rendered using the D3.js library [9]. The base map is generated with the *geoMercator* cylindrical projection, applied to country geometries provided in GeoJSON format. Each country feature is bound to a corresponding SVG element. Event listeners are registered for these elements, such that user interactions (e.g., country selection) trigger the rendering of the associated migration flow network on top of the map.

Migration flows are represented as quadratic Bézier curves connecting pairs of countries. Along each curve, a sequence of dashes and gaps is defined using the *stroke-dasharray* attribute. These paths are animated using D3's *transition* mechanism, creating the appearance of particles moving along the flow. The animation is implemented through a recursive *animatePath* function, which restarts the transition each time a dash-gap sequence completes. The animation continues as long as the path opacity remains nonzero. When a country is deselected, the opacity of the corresponding flows is set to zero, at which point the recursive animation terminates automatically. The SVG elements used to render the flows are then removed, preventing unnecessary resource usage and avoiding performance degradation when users interact with multiple countries sequentially.

Flow coloring is implemented using SVG *linearGradient* definitions. For each flow path, a corresponding gradient encodes the relative proportions of male and female migrants for the selected country and year. During hover interactions, the opacity of non-relevant flows is reduced while the opacity of flows associated with the focused countries is increased using D3 transitions. This interaction design highlights the relevant migration relationships and improves visual contrast.

4

COUNTRY-LEVEL STATISTICAL POP-UP

4.1 LEFT PANEL: TEMPORAL AND ECONOMIC CONTEXT

When a country is selected on the map, an auxiliary statistics panel is displayed to provide detailed temporal, economic, and demographic context for the migration flows shown in the main visualization. The panel is divided into two parts: a left panel focusing on temporal trends and economic indicators, and a right panel focusing on migration volume breakdowns by partner country and age group. All charts in this panel are dynamically updated based on the selected country, year, and migration direction (immigration or emigration).

4.1.1 • TIME SERIES OF MIGRATION VOLUME

The upper chart in the left panel displays a time series representing the total number of immigrants or emigrants associated with the selected country across all available years. A line chart with a linear time axis is used to encode this information, with time mapped to the horizontal axis and migration volume to the vertical axis.

A linear time series representation was chosen because it is well suited for showing temporal evolution, trends, and sudden changes in magnitude. This encoding allows users to easily identify long-term patterns, such as gradual increases or decreases in migration, as well as short-term anomalies or peaks. Compared to alternative encodings such as stacked bars or area charts, a simple line chart minimizes visual clutter and supports precise comparison between individual years.

The chart is linked to the global time slider: when a specific year is selected, the corresponding point on the time series is highlighted, reinforcing the temporal context of the currently displayed migration flows. When the play button is activated, the highlighted point moves along the timeline, visually synchronizing the animation of the flow network with changes in aggregate migration volume.

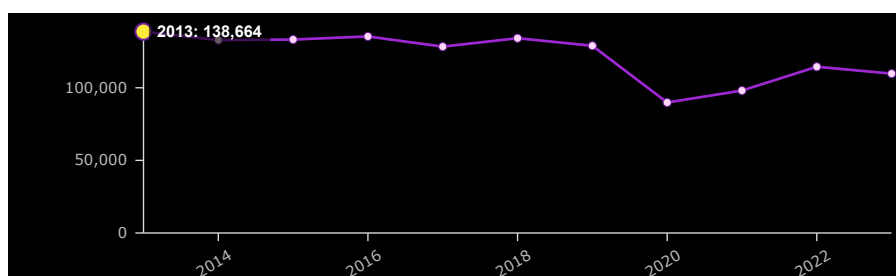


Figure 2: Time Series Volume for France from 2013 to 2023

4.1.2 • GDP BAR CHART

Below the time series, a horizontal bar chart displays the top ten countries by GDP for the selected year, including the selected country if it belongs to the top ten. Bars are ordered in descending order of GDP, enabling immediate comparison of economic scale between countries.

A horizontal bar chart was selected for this visualization because it supports efficient ranking and comparison across categories with long textual labels, such as country names. Horizontal orientation improves readability and avoids label overlap, which is especially important when displaying multiple countries simultaneously. The descending ordering further emphasizes relative differences in GDP, allowing users to quickly identify economically dominant countries in a given year.

This chart updates dynamically as the selected year changes, allowing users to observe how economic rankings evolve over time and how they relate to migration patterns. The GDP chart is intended to provide contextual information rather than causal explanation, helping users explore potential associations between economic size and migration flows.

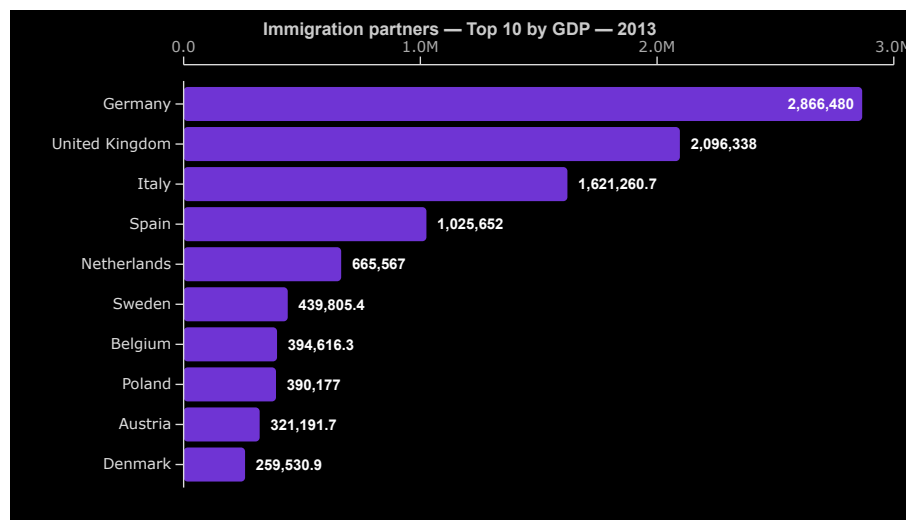


Figure 3: GDP of the top 10 partner countries for France

4.2 RIGHT PANEL: MIGRATION BREAKDOWN

The right panel provides a detailed breakdown of migration flows between the selected country and its partner countries. The visualization adapts depending on data availability, switching between a bar chart and a ridgeline plot.

4.2.1 • MIGRATION VOLUME BY PARTNER COUNTRY

When only aggregate migration data are available, the panel displays a horizontal bar chart showing migration volumes by partner country for the selected year and migration direction.

Countries are ordered in descending order of migration volume.

This representation allows users to quickly identify the most significant migration partners and to compare their relative importance. Horizontal bars were chosen for the same reasons as in the GDP chart: clear ranking, efficient use of space, and improved readability of country labels. In addition, interactive hovering is implemented in the bar chart: when the user hovers over an individual bar, the total migration volume for the corresponding country is displayed. The chart complements the flow network by providing precise quantitative values, which are more difficult to extract from animated flows alone.

4.2.2 • AGE GROUP BREAKDOWN USING RIDGELINE PLOTS

When age-disaggregated data are available, the bar chart transitions to a ridgeline plot as shown on Figure 4. In this representation, partner countries are arranged along the vertical axis, while age groups (from 0–4 to 85+) are mapped along the horizontal axis. For each country, the distribution of migrants across age groups is shown as a smoothed density curve with logarithmically scaled migration volume.

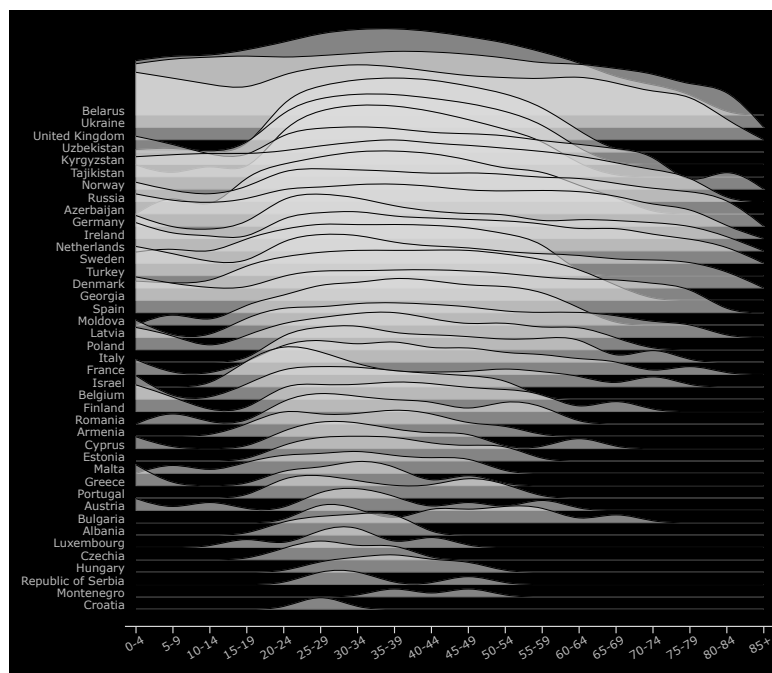


Figure 4: Age Group Breakdown for Lithuania, Immigration, 2023

Ridgeline plots were selected because they are particularly effective for comparing distributions across multiple categories while preserving a compact layout. This design allows users to simultaneously observe age-related migration patterns across several partner countries, such as differences in working-age migration or the presence of younger or older populations. Compared to grouped bar charts or stacked histograms, ridgeline plots reduce visual clutter and support pattern recognition across many distributions.

Interactive hovering is also implemented: when users hover over an individual ridgeline, the corresponding distribution is visually emphasized, and the total migration volume for that country is displayed as a tooltip. The transition between the bar chart and the ridgeline plot is animated, helping users maintain visual continuity and understand that both views represent different aspects of the same underlying data. As with other components of the panel, the ridgeline plot updates dynamically when the year or migration direction is changed.

5

DISCUSSION

The visualization design combines multiple visual encodings to support exploratory analysis of migration data while managing visual complexity. Animated flow networks are used to represent migration between countries, as they convey both directionality and magnitude more intuitively than static alternatives such as matrices or chord diagrams. Encoding migration volume through particle speed, rather than line thickness, reduces occlusion when multiple flows are present and maintains legibility across different spatial scales.

Demographic attributes are integrated directly into the flow representation through proportional color encoding. This approach allows gender distribution to be perceived immediately without introducing additional visual elements, preserving focus on the main spatial structure. In the statistical panel, horizontal bar charts are used to support ranking and precise comparison, while ridgeline plots are employed when age-disaggregated data are available, enabling comparison of distributions across multiple partner countries within a compact layout.

Interaction mechanisms are designed to support incremental exploration. The time slider and animation controls allow users to examine temporal variation without overwhelming the display, while coordinated updates across views maintain contextual consistency. The dynamic transition between different chart types reinforces the relationship between aggregate and disaggregated data.

Based on observations from the constructed visualizations, several notable migration patterns emerge. Migration flows are predominantly concentrated between neighboring countries, suggesting that geographic proximity plays a significant role in migration decisions. Gender composition appears largely balanced across most flows, with the ratio of male to female migrants remaining approximately equal in the majority of observed cases. Germany consistently emerges as the most attractive destination country in the region, which aligns with the economic context provided by the GDP visualization, where Germany also shows clear dominance. Additionally, a pronounced peak in immigration is observed in 2022 for countries such as Germany and Czechia. This increase corresponds to the recent regional conflict, with Ukraine identified as the primary source country driving these migration flows.

6

CONCLUSION

This project presents an interactive web-based visualization system for exploring European migration data using animated flow maps and coordinated statistical views. The system integrates spatial, temporal, demographic, and economic dimensions within a unified interface, enabling users to explore immigration and emigration patterns across countries and over time.

The visualization is implemented using D3.js library leveraging animation and interaction to reveal patterns that are difficult to observe in static representations. In particular, animated flow dynamics and coordinated temporal controls support intuitive exploration of complex migration relationships.

Overall, the project illustrates how interactive visualization techniques can enhance access to large-scale statistical data and support exploratory analysis of migration dynamics in Europe.

REFERENCES

- [1] Shalva Butkhuzi and Cesar Anibal Vilcherrez Zarate. Eurostat migration visualization. <https://shbutkhuzi.github.io/eurostat-vis/>, 2025. Interactive web-based visualization.
- [2] Eurostat. Eurostat database. <https://ec.europa.eu/eurostat/data/database>, 2025.
- [3] Eurostat. Immigration by age group, sex and country of previous residence (dataset code: migr_imm5prv). https://ec.europa.eu/eurostat/databrowser/view/migr_imm5prv/default/table?lang=en&category=migr.migr_cit.migr_immi, 2025.
- [4] Eurostat. Emigration by age group, sex and country of next usual residence (dataset code: migr_emi3nxt). https://ec.europa.eu/eurostat/databrowser/view/migr_emi3nxt/default/table?lang=en&category=migr.migr_cit.migr_emi, 2025.
- [5] Eurostat. International migration statistics metadata (euro sdmx metadata structure) – migr_immi_esms. https://ec.europa.eu/eurostat/cache/metadata/en/migr_immi_esms.htm#shortdata_descrDisseminated, 2025.
- [6] Shalva Butkhuzi and Cesar Anibal Vilcherrez Zarate. eurostat-vis: Visualization of eurostat migration data. <https://github.com/shbutkhuzi/eurostat-vis>, 2025. GitHub repository.
- [7] Eurostat. Gross domestic product (gdp) and main components (dataset code: nama_10_gdp). https://ec.europa.eu/eurostat/databrowser/view/NAMA_10_GDP/default/table, 2025.
- [8] Natural Earth. Natural earth: Free vector and raster map data. <https://nacis.org/initiatives/natural-earth/>, 2025.
- [9] Michael Bostock. D3.js — data-driven documents. <https://d3js.org/what-is-d3>, 2025.